

YIMIN TANG

(+1)4129547607 ◇ yimintan@usc.edu ◇ 3650 McClintock Ave, Los Angeles, CA, USA

Github: <https://github.com/TachikakaMin>, Website: <https://sites.google.com/view/yimintang>

EDUCATION

University of Southern California *August 2023 - Expected August 2027*

PhD student in Computer Science, Advisor: Sven Koenig, Erdem Bıyık
Viterbi School of Engineering

Carnegie Mellon University *August 2021 - August 2023*

Master of Science in Robotics, Advisor: Katia Sycara, Jiaoyang Li
Robotics Institute, School of Computer Science (RI, SCS)

ShanghaiTech University *September 2016 - July 2020*

B.E in Computer Science and Technology
The School of Information Science and Technology (SIST)

RESEARCH KEYWORDS

Multi-Agent Path Planning, Deep RL, Imitation Learning, Diffusion Policy, Foundation Model, World Model, LLM, Python, C++

WORK EXPERIENCE

Stripe, Machine Learning *May 2025 - August 2025*

Machine Learning Intern, mentored by Yiing Chau Mak

- Stripe Assistant with LLM

Futurewei Technologies, Machine Learning *May 2024 - May 2025*

Research Intern, mentored by Dr.Masood Mortazavi and Dr.Ning Yan

- Design new algorithms for LLM with long context about RAG and memory mechanism.
- 2024 Futurewei High Performance Intern Award

Microsoft, AzureStack Storage Team *August 2020 - August 2021*

Software Engineer, Full Time Engineer

- AzureStack is private cloud that provides same API as Azure and can be deployed on any small cluster.
- Designed and validated new features for Blob, Table, and Queue modules using C# and C++

OPEN SOURCE (PR MERGED)

PyTorch, Taichi, dreamer-torch, TrackFormer, Azure Java SDK, OI Wiki, onscripter-GBK-2015, MuJoCo, PlanViz, FFXIVChnTextPatch-Souma

SERVICE

Reviewer for WAFR, ICRA, IROS, ICAPS, CVPR, ICLR, ICML, NeurIPS, CoRL, AAAI, ECAI, AISTATS, JAIR, IEEE SMC, RA-L, TJSC

Organizer for The League of Robot Runners (LoRR) 2024 <https://www.leagueofrobotrunners.org/>
sponsored by Amazon Robotics

RESEARCH EXPERIENCE

IDM Lab, University of Southern California *Aug 2023 - Now*

Research Intern, mentored by Prof. Sven Koenig

LIRA Lab, University of Southern California *Aug 2024 - Now*

Research Intern, mentored by Prof. Erdem Bıyık

Multi-Robot Group, Carnegie Mellon University *Sept 2022 - Aug 2023*

Research Intern, mentored by Prof. Jiaoyang Li

- Used model-based method (World Model) to train Shared Modular Policies across multiple agents with diverse morphologies. This refined policy can adapt to new morphologies and tasks.
- Developed a communication program to facilitate data transfer across PyTorch, Unity ML-Agent API, Unity Mujoco plugin, and the Mujoco physical engine backend

AWARDS

Lambda's Research Grant Program
VESSL AI Support Program
2025 IEEE IES SYPA Travel Award
2024 Futurewei High Performance Intern Award
2023 USC Viterbi School of Engineering Fellowship
2023 CMU Algorithms With A Purpose (AWAP) AI Competition 2nd runner-up
2018 IBM China Outstanding Student Scholarship
2018 ACM-ICPC Xian Asia-East Continent Final Bronze Medal
2018 ACM-ICPC Asia Qingdao Regional Contest Silver Medal

PAPER LIST

Emerging Trends in Multi-Agent Path Finding from the League of Robot Runners
Daniel Damir Harabor, Zhe Chen, Yue Zhang, **Yimin Tang**, Sven Koenig, Jingjin Yu (**in review**)

Cube-Based Automated Storage and Retrieval Systems: A Multi-Agent Path Finding Approach
Yi Zheng **Yimin Tang**, Sven Koenig, T. K. Satish Kumar (**AAAI 2026** Workshop on Multi-Agent Path Finding, **in review**)

Judgelight: Trajectory-Level Post-Optimization for Multi-Agent Path Finding via Closed-Subwalk Collapsing
Yimin Tang, Sven Koenig, Erdem Biyik (**in review**)

RAILGUN: A Unified Convolutional Policy for Multi-Agent Path Finding Across Different Environments and Tasks
Yimin Tang*, Xiao Xiong*, Jingyi Xi, Jiaoyang Li, Erdem Biyik, Sven Koenig (**IROS 2025 Oral**, ICRA 2025 Workshop on Foundation Models, SoCS 2025 Extend Abstract)[PDF]

Accelerating Focal Search in Multi-Agent Path Finding with Tighter Lower Bounds
Yimin Tang, Zhenghong Yu, Jiaoyang Li, Sven Koenig (**IROS 2025 Oral**, RSS 2025 Workshop on Multi-Agent)[PDF]

Enhancing Lifelong Multi-Agent Path Finding with Cache Mechanism
Yimin Tang*, Zhenghong Yu*, Yi Zheng, T. K. Satish Kumar, Jiaoyang Li, Sven Koenig (**AAMAS 2025** Workshop on Autonomous Robots and Multirobot Systems, **in review**)

Enhancing Long Context Performance in LLMs Through Inner Loop Query Mechanism
Yimin Tang, Yurong Xu, Ning Yan, Masood Mortazavi (**NeurIPS 2024** Workshop on Adaptive Foundation Models)

ITA-ECBS: A Bounded-Suboptimal Algorithm for Combined Target-Assignment and Path-Finding Problem
Yimin Tang, Sven Koenig, Jiaoyang Li (**SoCS 2024**)[PDF]

Solving Multi-Agent Task Assignment and Path Finding with a Single CBS Tree
Yimin Tang, Zhongqiang Ren, Jiaoyang Li, Katia Sycara (**IEEE MRS 2023**, **Best Paper Candidate**, also presented at ICAPS 2024)

Multi-modal perception and transfer learning for grasping transparent and specular objects
Thomas Weng, Amith Pallankize, **Yimin Tang**, Oliver Kroemer, David Held (**IEEE RA-L** presented at ICRA 2020) [WEB]